DIGITAL IMAGE FUNDAMENTALS

Understanding Pixel Representation, Spatial Resolution, Intensity Resolution, and Grayscale Images

TECHNICAL PRESENTATION

CO1-AT3 ITA 0515

PRESENTED BY

VARSHA R 192421045

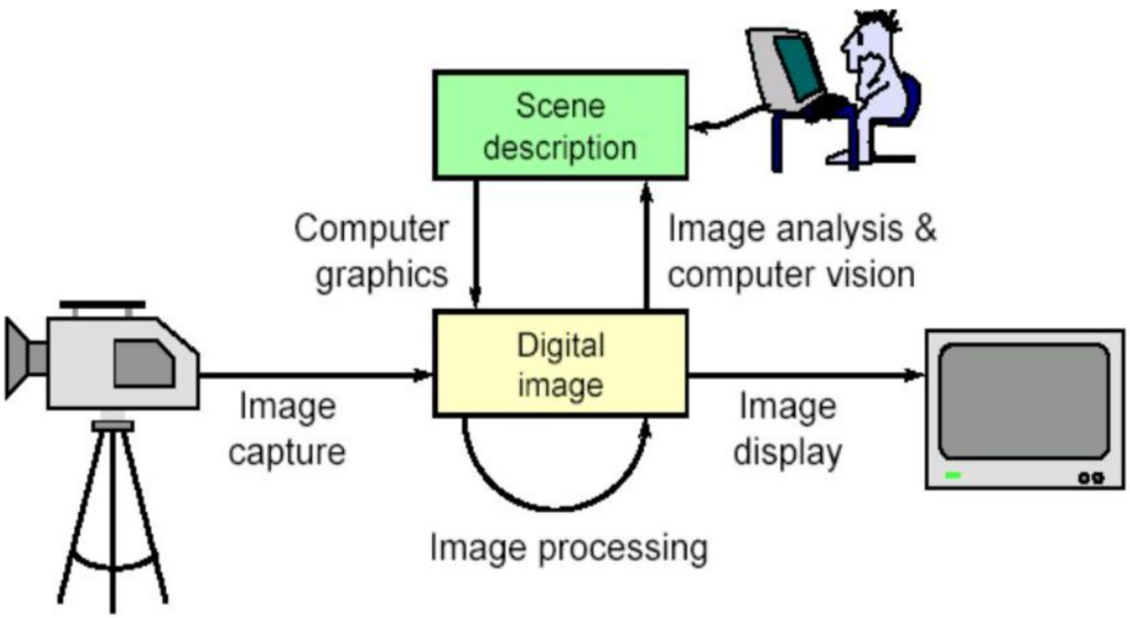
VIDDHIYASHREE 192421188

K Sreelatha 192425248

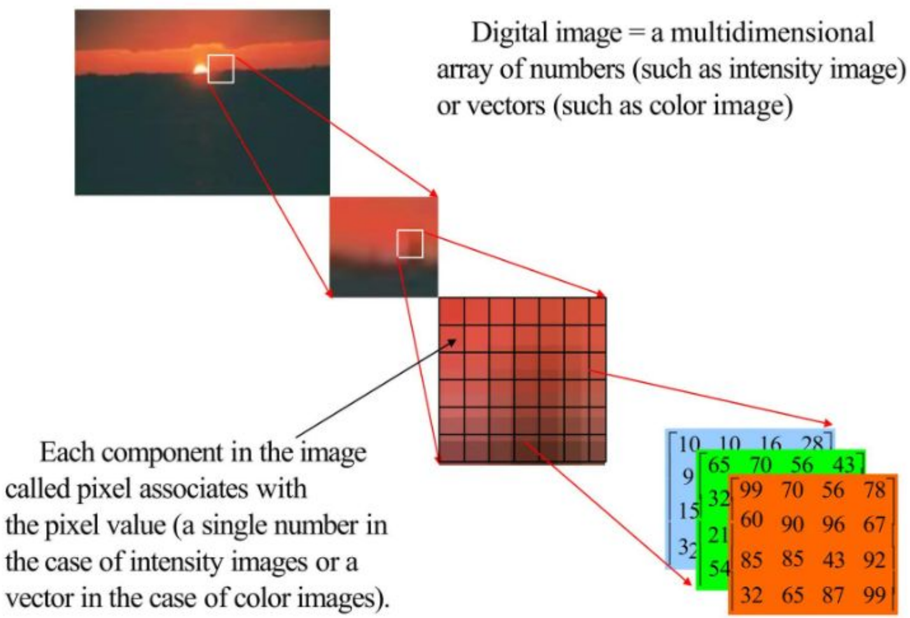
K Amrutha 192425366

INTRODUCTION

- - A digital image is a visual representation stored in digital form.
- - It consists of pixels arranged in rows and columns.
- - Used in photography, medical imaging, satellite imaging, and computer vision.
- - Image quality depends on resolution and intensity levels.



OBJECTIVES



- - Understand digital image fundamentals.
- - Explain pixel representation.
- - Study spatial and intensity resolution.
- - Understand grayscale images.
- - Identify common image quality issues.

PROBLEM STATEMENT

- - Low spatial resolution causes blurred images.
- - Insufficient intensity resolution reduces detail.
- - Noise and distortion affect clarity.
- - Poor image quality impacts analysis and decision making.

Problem: How can image quality be represented and maintained effectively

DIGITAL IMAGE FUNDAMENTALS

- Pixel Representation: Smallest unit of an image.
- Spatial Resolution: Number of pixels in an image.
- Example: 1920 × 1080.
- Intensity Resolution: Number of brightness levels.
- Example: 8-bit = 256 gray levels.
- Grayscale Image: Values range from 0 (black) to 255 (white).

- Picture elements (pixels)
  - Sampling, quantization
- Higher dimensional image -- voxels
- Bi-level images (black/0 or white/1)
- Grayscale images
  - 1 byte/pixel: 256 gray levels

- Color images
  - True color: RGB 24bits/pixel

- Image size, e.g. VGA 640x480
  - Grayscale image: 307,200 bytes
  - True color image: 921,600 bytes

CONCLUSION

- - Pixels form digital images.
- - Spatial resolution determines image detail.
- - Intensity resolution determines brightness accuracy.
- - Grayscale images are important in image processing.
- - These concepts are fundamental to computer vision.

Thank you!